

Name: _____



Homework Questions for Climate and Us Week 1

Due the week of November 17

SHOW ALL WORK

1. Earth's Energy Balance

Any given planet's climate is determined by its energy balance. Earth's energy balance is such that it creates a livable environment for humans and other life, unlike those of nearby Venus or Mars. Climate forcings are factors that drive changes in climate and alter the Earth's energy balance. Use Figure 1 (below) to answer the following questions.

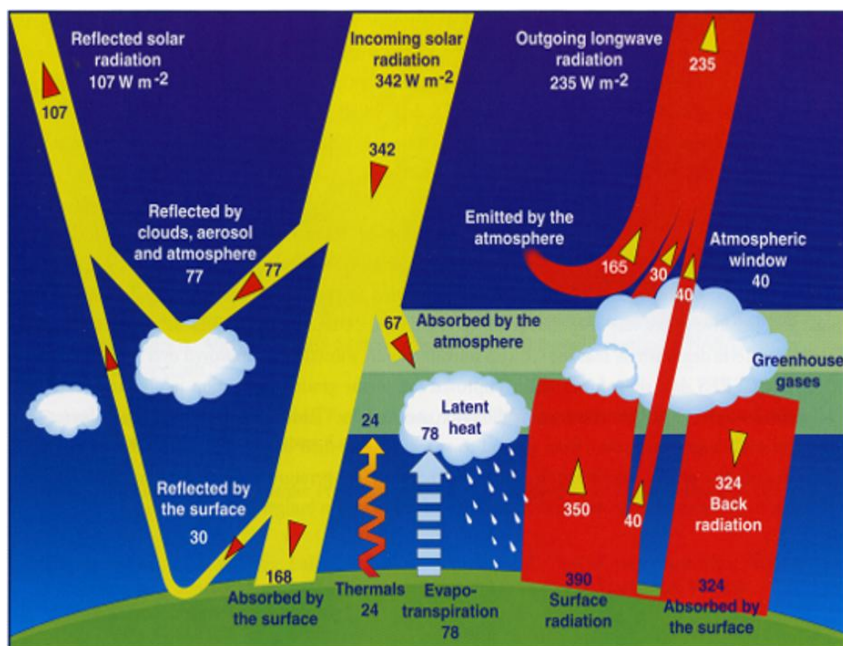


Figure 1. Energy balance diagram showing the flow of energy (in Watts per square meter, or W m^{-2}) from the Sun, from Earth's surface, and through the atmosphere. Yellow regions show incoming solar (shortwave) radiation. Red regions show outgoing infrared (longwave) radiation.

1A. (i) What three values illustrate the balance between incoming and outgoing energy? Set up an equation with these three values illustrating this balance.

1A. (ii) Where does the incoming solar radiation go? Name the possible outcomes for incoming solar radiation based on the diagram.

1A. (iii) What values illustrate the balance between energy absorbed at the surface and lost at the surface? Select all that apply and demonstrate the two energy flows are equal with a calculation.

Energy absorbed at the surface:

- a. Solar radiation: 168 W/m^2
- b. Thermals: 24 W/m^2
- c. Surface radiation: 390 W/m^2
- d. Back radiation: 324 W/m^2
- e. Evapotranspiration: 78 W/m^2

Energy lost at the surface:

- a. Solar radiation: 168 W/m^2
- b. Thermals: 24 W/m^2
- c. Surface radiation: 390 W/m^2
- d. Back radiation: 324 W/m^2
- e. Evapotranspiration: 78 W/m^2

1A. (iv) The outgoing longwave radiation is made up of energy emitted by the atmosphere (165 W/m^2) plus energy emitted by the clouds (30 W/m^2), plus energy escaping through the atmospheric window (40 W/m^2).

What components together account for the energy emitted by the atmosphere plus that emitted by the clouds? Select all that apply. *Hint: think about which components need to be added OR subtracted.*

- a. Energy absorbed by the atmosphere coming in: 67 W/m^2
- b. Thermal energy to the atmosphere: 24 W/m^2
- c. Evapotranspiration to make clouds: 78 W/m^2
- d. Energy reflected by the surface: 30 W/m^2
- e. Surface radiation (not including atmospheric window): 350 W/m^2
- f. Surface radiation (including atmospheric window): 390 W/m^2
- g. Back radiation: -324 W/m^2

1B. (i) How would an increase in albedo, defined as the amount of shortwave energy reflected by the Earth (surface + atmosphere), affect the energy balance in Figure 1? Write 'increase', 'decrease', or 'not change' in the provided blanks.

- a. Shortwave energy absorbed by Earth's surface (168 W/m^2) would _____
- b. Longwave energy input to the atmosphere by surface radiation (350 W/m^2) would _____
- c. Incoming solar radiation (342 W/m^2) would _____
- d. Earth's average surface temperature would _____

1B. (ii) Table 1 below gives the value of albedo for various surfaces. **Which of the following would cause a decrease in Earth's surface albedo?** Select all that apply.

Substance or Surface	Albedo
Whole Earth average	0.3
Fresh snow	0.8-0.9
White paint	0.5-0.9
Sea ice	0.5-0.7
Desert sand	0.4
Green grass	0.25
Bare soil	0.17
Dark roofs	0.1-0.2
Conifer forest	0.08-0.15
Open ocean	0.06
Fresh asphalt	0.04

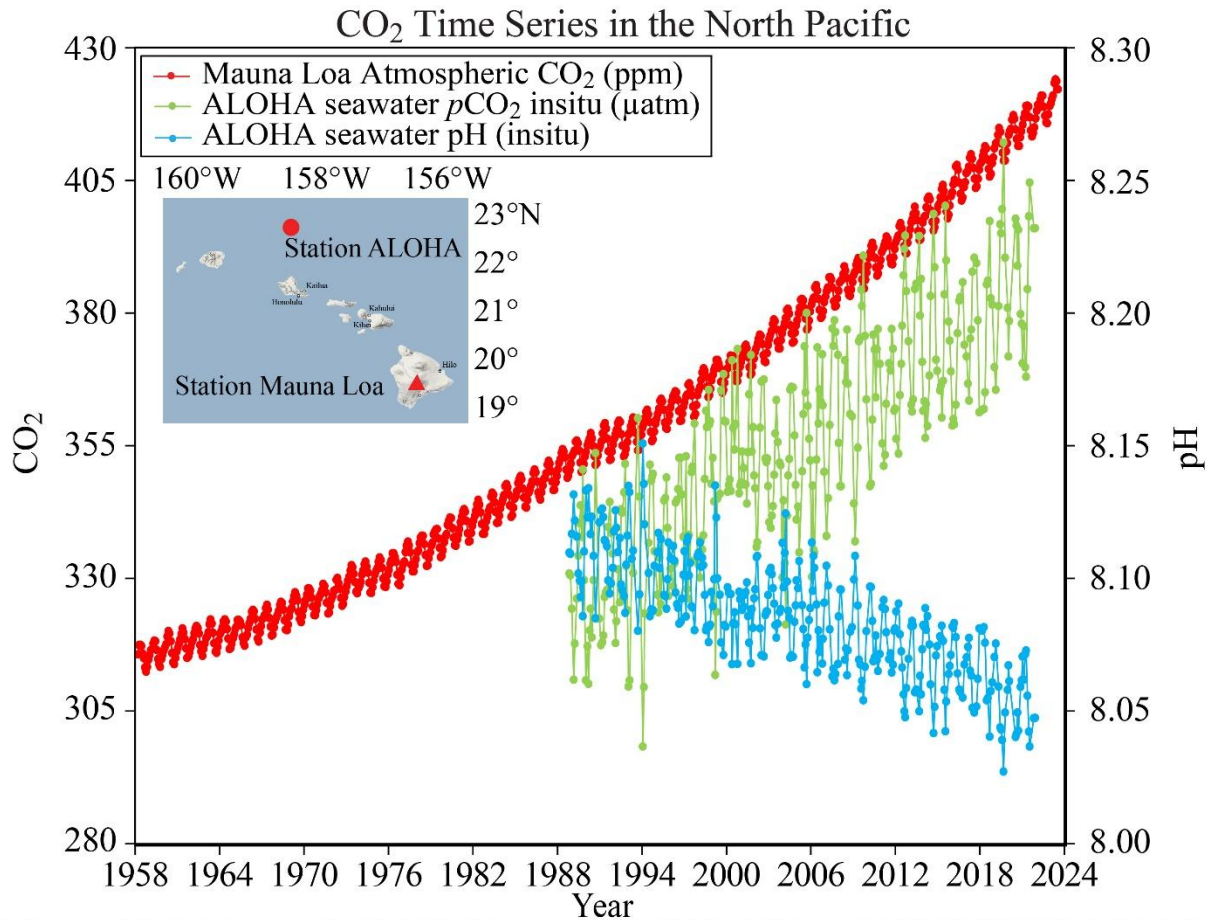
Table 1. Albedo of various substances and surfaces, defined as the fraction of solar radiation reflected. (Data from Akbari et al., 2012)

- a. Allowing the size of the Sahara Dessert to increase and take over forests
- b. Melting sea ice
- c. Replacing a grassy field or forested park with a parking lot
- d. Planting grass in open fields of bare soil
- e. Sea level rise increasing the surface area of Earth's oceans
- f. Painting roofs of buildings white

1C. TRUE OR FALSE (correct the statement if false): Greenhouse gases largely absorb incoming solar (shortwave) radiation.

2. Recent Climate Change and Impacts

Figure 2 (next page) shows measurements of atmospheric CO₂ from the Mauna Loa Observatory station in Hawaii starting in 1958. Seawater pH and seawater partial pressure CO₂ have also been measured at the ALOHA station off the coast of the Hawaiian island O'ahu (see inset map) since approximately 1990, as shown in Figure 2.



Data: Mauna Loa (https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_mm_mlo.txt) ALOHA (https://hahana.soest.hawaii.edu/hot/hotco2/HOT_surface_CO2.txt) ALOHA pH & pCO₂ are calculated at in-situ temperature from DIC & TA (measured from samples collected on Hawaii Ocean Times-series (HOT) cruises) using co2sys (Pelletier, v25b06) with constants: Lueker et al. 2000, KSO4: Dickson, Total boron: Lee et al. 2010, & KF: seacarb

Figure 2. Measurements of atmospheric CO₂ concentration (red line) from the Mauna Loa Observatory station from 1958 to 2024, seawater partial pressure of CO₂ in μatm (green line) and seawater pH (blue line) from station ALOHA from 1990 to February 2024. Source: National Oceanic and Atmospheric Administration Pacific Marine Environmental Laboratory Carbon Program.

2A. The record of atmospheric CO₂ concentrations from Mauna Loa, Hawaii shows both a long-term trend from 1958-2024, as well as short-term (intra-annual) variability. **Explain what causes intra-annual variability in atmospheric CO₂ in 2-3 sentences.**

2B. (i) Using Figure 2, estimate the annual rate of change for atmospheric CO₂ concentration (in ppm CO₂ per year) at the Mauna Loa Observatory station for the time period from 1958-1976. Show your work, then select the multiple choice answer that is closest to the value you calculated.

- a. 0.05 ppm/year
- b. 0.1 ppm/year
- c. 0.8 ppm/year
- d. 1.6 ppm/year
- e. 2.3 ppm/year

2B. (ii) Now estimate the average annual rate of change for the time period from 2006 to 2024 (Column A). How many times larger is this annual rate of change in atmospheric CO₂ compared to the period from 1958-1976, roughly half a century earlier (Column B)? Show your work, then select the multiple choice answers that are closest to the values you calculated. Note: use the multiple choice value selected in Column A to determine the rate for Column B.

- A
- a. 0.05 ppm/year
 - b. 0.1 ppm/year
 - c. 0.8 ppm/year
 - d. 1.6 ppm/year
 - e. 2.3 ppm/year

- B
- f. It is the same rate
 - g. The 2006-2024 rate is 1.5 times larger
 - h. The 2006-2024 rate is 3 times larger
 - i. The 2006-2024 rate is 4 times larger
 - j. The 2006-2024 rate is 5 times larger

2C. As the concentration of CO₂ in the atmosphere increases, the concentration of CO₂ dissolved in seawater also increases, causing a change in pH. Based on Figure 2, what can we conclude about the relationship between atmospheric CO₂ concentration and seawater acidity? Note that a decrease in pH means the water is becoming more acidic while an increase in pH means the water is becoming more basic.

2D. The ocean is a major carbon “sink” and currently absorbs about 31% of the CO₂ released by anthropogenic sources into the atmosphere (Gruber et al., 2019). This means the rate you calculated in question 2B is actually only 69% of what it would be if the ocean were not able to absorb atmospheric CO₂. Unfortunately, the ocean’s ability to absorb CO₂ is finite, and has been declining over the past decade as warmer oceans have less capacity for CO₂ uptake. **Based on your multiple choice answer to 2B (ii), what would the concentration of atmospheric CO₂ be this year (2025) if ocean absorption of atmospheric CO₂ had all of a sudden gone from 31% per year to 0% per year starting in 2006 (when atmospheric CO₂ was 380 ppm)?** Compare this value to the observed atmospheric CO₂ that we have today of 425 ppm (Oct 2025 average).

2E. Figure 3 shows the flux of carbon (in petagrams of C per year) going into the atmosphere measured at the Mauna Loa Observatory in Hawaii (blue line), in ice core bubbles from Antarctica (black line) and released by the burning of fossil fuels (red line).

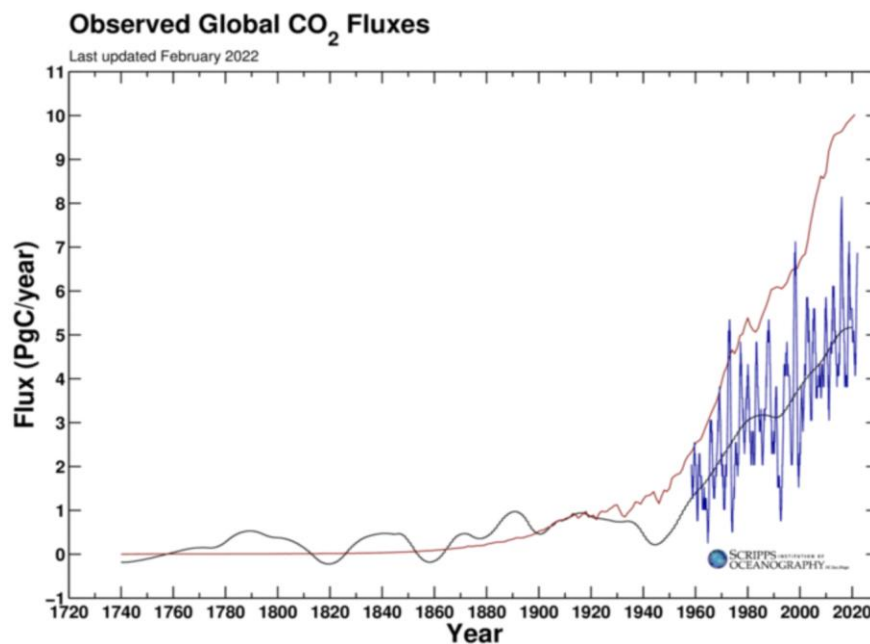


Figure 3. The annual flux of CO₂ (mass per year) is calculated using the yearly increase in atmospheric CO₂ concentration. Data from Scripps Institute of Oceanography.

2E. (i) Based on Figure 3 (previous page), estimate the difference between the flux of carbon released by fossil fuels and the flux of carbon measured in Antarctica in 2020. Give your answer in kilograms of carbon per day and write your answer in scientific notation (1 petagram = 1×10^{15} grams).

2E. (ii) We already learned that some of the carbon released by the burning of fossil fuels is being absorbed by the ocean. Where else could it be going? **Provide another reasonable carbon sink on Earth where the excess carbon you calculated in 2E. (i) is going.**

2F. In Prof. Helfand's lecture, you learned about the use of isotope ratios to discern the source of atmospheric carbon. Figure 4 below shows the ratio of $^{13}\text{C}/^{12}\text{C}$ in atmospheric CO_2 (relative to a standard), which is denoted as $\delta^{13}\text{C}$. The black line shows the value recorded at the Mauna Loa Observatory, Hawaii, and the red line shows the value recorded at the South Pole, Antarctica.

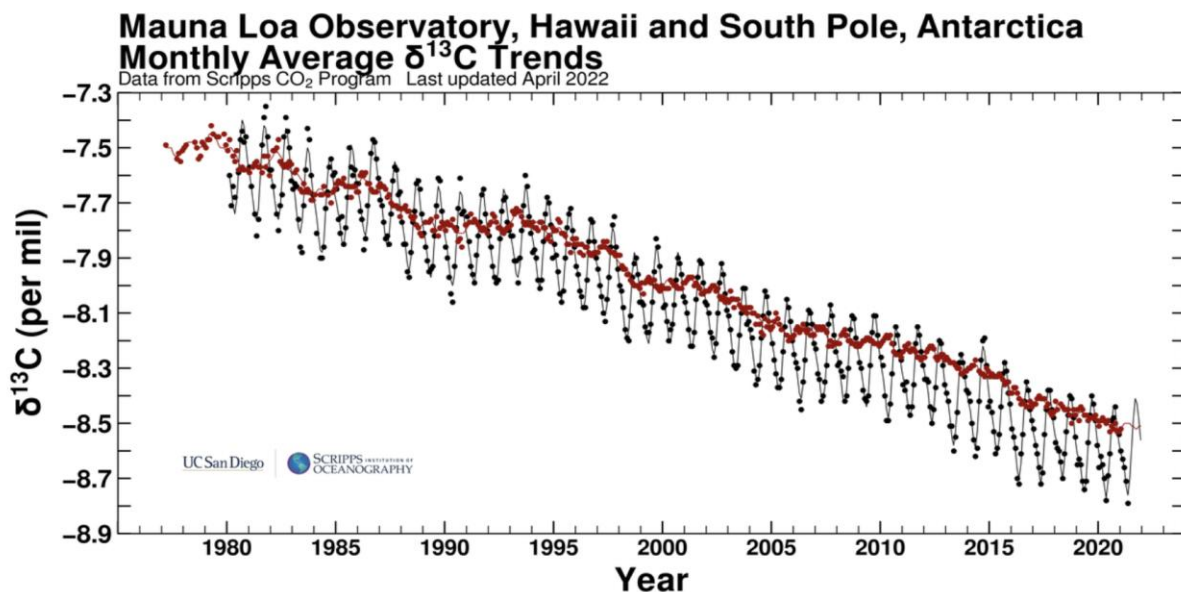


Figure 4. Average monthly $\delta^{13}\text{C}$ records of atmospheric CO_2 from Mauna Loa Observatory, Hawaii (black dots/curve) and South Pole, Antarctica (red dots/curve) from 1975 to 2022. (Note: $\delta^{13}\text{C}$ represents the ratio of $^{13}\text{C}/^{12}\text{C}$ relative to a standard and is recorded in parts per thousand or “per mil”). Source: Scripps CO_2 program.

2F. (i) Changes in $\delta^{13}\text{C}$ represent the ratio of $^{13}\text{C}/^{12}\text{C}$ relative to a standard, so a higher (less negative) $\delta^{13}\text{C}$ corresponds to a higher $^{13}\text{C}/^{12}\text{C}$ ratio and a lower (more negative) $\delta^{13}\text{C}$ corresponds to a lower $^{13}\text{C}/^{12}\text{C}$ ratio. **Based on what you learned in lecture, why does a decline in $\delta^{13}\text{C}$ through time provide support that fossil fuels are the source of increasing atmospheric CO_2 ?** Explain in 2-3 sentences.

2F. (ii) Which location shows greater intra-annual variability in $\delta^{13}\text{C}$ value of atmospheric carbon dioxide: the South Pole, Antarctica or Mauna Loa, Hawaii? Circle the correct answer and in 2-3 sentences, speculate as to what may be responsible for this difference.